

CNN-aided morphological classification of radio galaxies

A (partial) replication study

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Abstract

This is a partial replication study of the work on neural network-aided morphological classification of radio galaxies by Brand et al. (2023) [1]. The original paper investigated two approaches to improve the classification accuracy of CNN models trained to tackle such problem: the first proposal consists in applying preprocessing and augmentation techniques to the images before training the network; the second consists in introducing feature-guided network architectures.

This work focuses solely on the replication of the evaluation of the preprocessing techniques. Particularly, two strategies are compared by training copies of the network on three variations of the original authors' FRGMRC dataset: a baseline variant with only pixel-normalization applied, one with PCA-based rotational standardization applied to each sample, and one where each training sample is augmented with rotated copies.

The approach by Brand et al. was replicated with some changes to the network architecture, and the results obtained were found to be consistent with the trend presented in the original paper, though with differences in the magnitudes. These differences can be explained by the modified CNN architecture, as well as by other slight variations from the original approach, which are all documented and discussed.

1 Introduction

The classical radio galaxy taxonomy distinguishes Fanaroff-Riley type I (FRI) and type II (FRII) sources based on the ratio of the distance between their 'hot spots' to the total distance between the edges of the radio source; alongside compact (which

are referred to as 'Compact' galaxies) and bent-tailed morphologies (which are referred to as 'Bent'). For more information regarding this taxonomy, readers are redirected to the reference paper by Brand et al. [1], and to the original work on radio galaxies by Fanaroff and Riley [2].

With the increase in the volume of data produced by radio telescopes, convolutional neural networks (CNNs) have become an essential tool to speed up the morphological classification of radio galaxies. Despite the ability to automate the task, their application requires dealing with a key challenge: the extreme rotational variability of the sources. The same galaxy observed from different angles can appear as different pixel patterns, thus greatly increasing the intra-class variance and making the classification harder for a model trained on such orientation-biased data.

Brand et al. [1] proposed two strategies to address the problem: the first is to act on the data before feeding it to the network, while the second consists in implementing feature-guided architectures that can leverage domain-knowledge to improve the training process. Particularly, the proposed preprocessing techniques are PCA-based rotational standardization, aimed at standardizing the rotation of all samples by aligning the galaxy's principal component to the main image axis; and rotational augmentation, to produce multiple rotated samples of each training item.

The aim of this work is to replicate the preprocessing approach to evaluate the effects of the two techniques on the classification accuracy and training times of the employed neural network. To do so, multiple copies of a simple CNN architecture are trained on three variants of the original dataset. The resulting models are then evaluated and their performances are compared. Finally, the results are compared to those reported in the original paper to draw conclu-

sions about the efficacy of the proposed solutions.

Section 2 provides an overview of the dataset, the employed methods and how they have been replicated, the network architecture and the differences with that proposed by Brand et al, and the training protocol. Section 3 presents the results of the experiment (3.1) and a comparison with those reported in the original study (3.2). Finally, Section 4 reports closing observations and draws conclusions.

2 Method Overview

2.1 Dataset

For the replication study, the same dataset used by the original authors was utilised. Particularly, it is the FRGMRC dataset [3], consisting of 960 radio galaxy images of 300×300 pixels labelled into four morphological classes. Table 1 shows the class composition of this dataset.

Table 1: Class composition of the dataset.

Class	Samples
Compact	208
FRI	182
FRII	357
Bent	213
Total	960

In this work, all 960 images are used (whereas the original paper excluded a subset of 92 sources after manual inspection), and they are split into training (80%), validation (10%) and test (10%) sets, resulting in 768, 96 and 96 images respectively. The proportions of these splits are shown in 2. Furthermore, the frequencies of classes in each split are representative of their proportions in the complete dataset, as to not introduce bias in the training and evaluation phases.

Table 2: Number of samples in training, validation, and test data set.

Training	Validation	Test	Total
768	96	96	960

2.2 Preprocessing strategies

For the study, all the preprocessing functions have been implemented in MATLAB according to their descriptions in the original paper. While a short summary is provided below, readers are referred to [1] for

the algorithmic details, and to the authors’ repo [4] for the original implementation. In short, the employed strategies are the following:

Normalization All images are normalized with min-max scaling to the range $[0, 1]$, following Equation (1) of the original paper. This step is applied to every sample in all configurations.

Center crop A 150×150 central crop is applied to each normalized image to match the crop used in the original implementation.

Thresholding For each image, a binary mask identifying the galaxy pixels is computed by an adaptive thresholding algorithm. The algorithm distinguishes between low-noise and high-noise images based on the histogram distribution of pixel intensities, and applies morphological operations (dilations, small-component removals, hole fillings) to obtain a clean mask. Note that the mask is used only as input to the rotational standardization step, and is not applied to the image that is fed to the network.

Rotational standardization As explained in Section 2.2.4 of [1], the P pixel coordinates of the thresholded mask are centered around their mean, arranged into a $2 \times P$ matrix X , and decomposed as $X = U\Sigma V^\top$ using Singular Value Decomposition (SVD). A rotation angle is then extracted from U , and the original normalized image is rotated by this angle, aligning the principal axis of the galaxy with the horizontal axis of the image.

Rotational augmentation As detailed in Section 2.2.3 of the original paper, the training set is augmented by generating five additional rotated copies of each image, at angles of 60° , 120° , 180° , 240° and 300° with respect to the original sample.

These strategies are utilised to obtain three variants of the original dataset to train and evaluate the CNNs. First, normalization is applied to all samples to produce a baseline variant. Then, from this baseline, a standardized variant and an augmented variant are obtained by applying rotational standardization and augmentation, respectively. More precisely, rotational standardization is applied to image samples of all three splits; while augmentation is solely used to produce an augmented training set, as applying it to the test set would lead to an unfair comparison of the two approaches.

2.3 Network architecture

The network employed for this study is a CNN that follows the general SCNN architecture described in Section 3.1 of the original paper[1]: it consists of three convolutional blocks, a flatten layer, two dense layer blocks, and a final softmax layer.

Despite employing the same general organization, the network used here differs from the original in two main aspects:

1. **Convolutional blocks** - The original SCNN uses a first convolutional layer with 64 filters and a 14×14 kernel, followed by two convolutional blocks with 7×7 kernels and 128 and 256 filters, respectively. This implementation adopts smaller, fixed-size 3×3 kernels throughout the whole network, and reduces the number of filters to 32, 64, and 128 in the three blocks.
2. **Fully connected layers** - The original network uses two dense layers with 150 and 75 units, respectively, both activated by ELU functions. Here, the dense layers have 256 and 128 units and use ReLU activations.

The choice of ReLU over ELU was made to avoid the implementation of a custom layer, since MATLAB does not provide ELU natively. This simplified the network implementation and, even though it may have influenced the absolute performance results, it should have had no effect on the relative rankings of the preprocessing strategies. Furthermore, the use of smaller kernels and fewer filters allows to understand whether the preprocessing techniques remain effective even when paired with a less expressive architecture.

Table 3: Layer configuration of the SCNN variant.

Block	Layer(s)	Output shape
Input	—	$150 \times 150 \times 1$
Conv 1	2× Conv 3×3 , 32 MaxPool 2×2	$75 \times 75 \times 32$
Conv 2	2× Conv 3×3 , 64 MaxPool 2×2	$37 \times 37 \times 64$
Conv 3	2× Conv 3×3 , 128 MaxPool 2×2	$18 \times 18 \times 128$
Flatten	—	41472
Dense 1	FC 256 + ReLU + Dropout 0.5	256
Dense 2	FC 128 + ReLU + Dropout 0.5	128
Output	FC 4 + Softmax	4

Table 3 highlights further details of the SCNN variant used in this work, such as pooling layers and dropout, and shows the output shape of each block.

2.4 Training protocol

A different copy of the above discussed CNN is initialized and trained on each of the three dataset variants to perform an experiment. Particularly, each experiment is repeated 20 times with independent random initializations: Glorot for convolutional layers, and He for dense layers. The final reported metrics are the averages over these runs.

All configurations are trained for a maximum of 100 epochs, with early stopping if the validation loss does not improve for 5 consecutive epochs. Categorical cross-entropy loss is used to train all network instances. A constant learning rate of 10^{-4} is used, and the mini batch size is 32.

The only notable difference with the original training setup is that the Adam optimizer is used in place of Nadam. Nevertheless, this should not influence the relative performances of the approaches.

3 Results

3.1 Experiment results

Table 4 reports the average performance of the SCNN across 20 independent runs on each dataset, along with standard deviation values. The ranking anticipated by the original paper is reproduced: not performing any kind of preprocessing on the images leads to the worst macro F1 score (0.894 ± 0.017), while both rotational standardization and augmentation bring performance improvements. In particular, augmentation leads to a higher macro F1 (0.927 ± 0.013), compared to the standardized configuration (0.913 ± 0.018).

Though the original trend is replicated, the magnitude of the improvements appears to be of smaller scale. Thus, to verify that the observed ranking is consistent across runs, and not the product of random variabilities, Welch two-sample t -tests between all pairs of configurations are performed with the Bonferroni-corrected significance level $\alpha = 0.05/3 \approx 0.0167$. Results of these tests are shown in table 5: all differences are found to be statistically significant, confirming that, indeed, rotational standardization and augmentation both lead to improvements in classification accuracy.

What is more, it is important to note that the improvements in classification accuracy are accompanied by substantial differences in training costs. Rotational standardization reduces the average training time by a factor of ≈ 2.45 with respect to the baseline (56 s vs 138 s), whereas augmentation increases it by a factor of ≈ 3.19 (441 s vs 138 s). The training

Table 4: Average performance of SCNNs over 20 independent runs. Values reported as mean $\pm$ std.

Pre-processing	Average macro F1	Average final validation loss	Average training time (s)
Baseline	0.894 ± 0.017	0.495 ± 0.057	138.4 ± 22.7
Standardization	0.913 ± 0.018	0.424 ± 0.049	56.4 ± 8.3
Augmentation	0.927 ± 0.013	0.340 ± 0.076	441.3 ± 72.4

Table 5: Two-sample t -tests on the macro F1 distributions, with Bonferroni-corrected significance threshold: $\alpha = 0.05/3 \approx 0.0167$.

Comparison	t	p -value	Significant
Base vs Std	-3.30	2.09×10^{-3}	Yes
Base vs Aug	-6.78	7.74×10^{-8}	Yes
Std vs Aug	-2.86	7.21×10^{-3}	Yes

cost of augmentation is therefore almost $8\times$ that of standardization, while the improvement in macro F1 over standardization is only $+0.014$. This trade-off is consistent with the original paper and suggests that standardization is preferable in resource-constrained scenarios.

Table 6 breaks down the results by morphological class and a few observations emerge. First, the Compact class is the easiest to classify across all configurations, consistent with its compact and symmetric morphology. Second, augmentation yields the best performance improvements on Compact and FRI galaxies, while achieving slightly worse performance than rotational standardization on the Bent class. Finally, both techniques only have marginal effects on the identification of the FR II class, suggesting that the network cannot gain much extra relevant information from either approach.

Furthermore, precision for the FR II class saturates at 0.997 with both strategies, but recall decreases from 0.907 (baseline) to 0.868 (standardization) and 0.869 (augmentation). In fact, the confusion matrices in Figure 1 show that both rotational standardization and augmentation lead to an increase in the misclassifications of the FR II class. This suggests that the network, when uncertain, is biased by the preprocessing towards predicting Bent or FRI rather than FR II.

3.2 Comparison with original results

As presented in Section 3, the qualitative trend reported by Brand et al. [1] is reproduced. However, the magnitudes of the improvements differ substantially

from those of the original study, and this discrepancy deserves a dedicated discussion.

Table 7 reports a comparison between the macro F1 values reported in the original paper and those obtained in this replication study, highlighting two key points. First, the gains obtained from both preprocessing strategies are considerably smaller: the standardization improvement is $+0.019$ against $+0.074$ in the original paper, and the augmentation improvement is $+0.033$ against $+0.095$. As a result, the performance gap between augmentation and baseline is approximately three times smaller here (0.033 vs 0.095). Second, the baseline performance is slightly higher here (0.894 vs 0.876).

The most plausible explanation for the reduced magnitude lies in the architectural difference. The network architecture chosen for this replication study (see Section 2.3) is simpler and less expressive than the original SCNN: it uses 3×3 convolution kernels instead of the 7×7 and 14×14 ones, as well as employing fewer filters per layer (32/64/128 instead of 64/128/256). A less expressive model has a smaller capacity to exploit the additional variability introduced in the training data, which would explain why the benefit of augmentation, that is, the strategy that increases the effective size of the training set the most, shrinks more than the benefit of standardization.

This interpretation is supported by two observations. First, the augmented training set is larger in our setup than in the original ($777 \times 6 = 4662$ images vs $702 \times 6 = 4212$), yet the augmentation gain is smaller: if the benefit depended only on the amount of additional data, the opposite would be expected. Second, the higher baseline accuracy in our setup (0.894 vs 0.876) suggests that the simpler model hit a sweet spot in the level of complexity that allowed it to generalize well without any rotational preprocessing, leaving less room for improvement.

Furthermore, the training time comparisons reveal an even more pronounced discrepancy. In the original paper, standardization is only ≈ 1.2 times faster than the baseline (98 s vs 116 s), while in this experiment the speed-up is much larger, at approximately $2.45\times$ (56 s vs 138 s). Conversely, the difference in the costs of augmentation across the two studies appears less

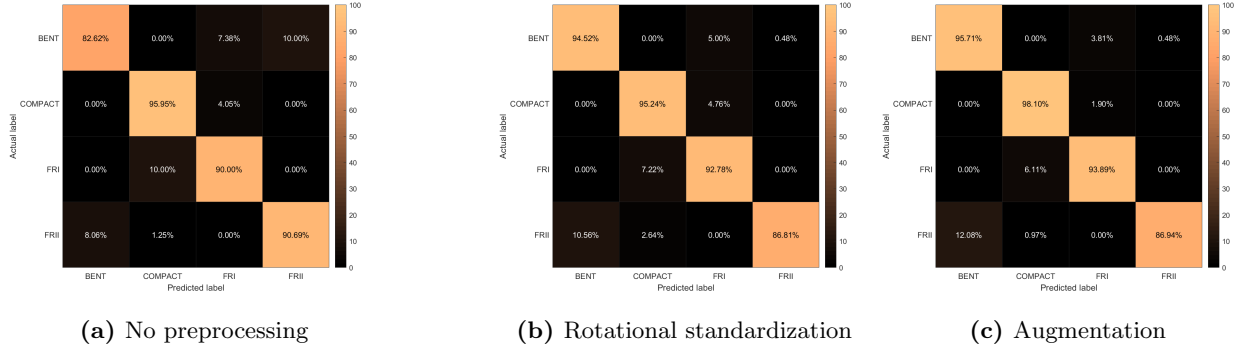


Figure 1: Average confusion matrices over 20 independent runs for the three preprocessing strategies. Values are row-normalized percentages.

Table 6: Average per-class performance of SCNNs over 20 independent runs. Values are reported as mean over runs.

Pre-processing	Bent			Compact			FRI			FRII		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Baseline	0.860	0.826	0.840	0.902	0.960	0.929	0.873	0.900	0.885	0.941	0.907	0.923
Standardization	0.843	0.945	0.890	0.900	0.952	0.925	0.891	0.928	0.908	0.997	0.868	0.927
Augmentation	0.824	0.957	0.885	0.935	0.981	0.957	0.935	0.939	0.937	0.997	0.869	0.928

Table 7: Comparison of macro F1 gains with the original paper.

	Original		Replication	
Base	0.876		0.894	
Std	0.950	+0.074	0.913	+0.019
Aug	0.971	+0.095	0.927	+0.033

Table 8: Comparison of training times and training speedup with the original paper.

	Original	Replication
Base	116	138.4
Std (speedup)	98 ($\times 1.18$)	56.4 ($\times 2.45$)
Aug (speedup)	476 ($\times 0.24$)	441.3 ($\times 0.31$)

harsh: $\approx 4.1\times$ slower than the baseline in the reference paper (476s vs 116s), and $\approx 3.2\times$ here (441s vs 138s). The stronger speed-up of standardization may be partly attributed to the larger training set size: since the baseline here already requires more time to converge, the reduction in intra-class variability introduced by standardization translates into larger absolute and relative gains.

At the per-class level, it is interesting to notice that both studies report significant improvements in the

F1 scores for the Bent class when standardization is applied (+0.187 in the original, +0.050 here), for the Compact class when augmentation is applied (+0.032 in the original, +0.028 here), and for the FRI class when augmentation is applied (+0.057 in the original, +0.052 here). Conversely, the improvement in the F1 score for the FRII class is much less impressive here (+0.005 with augmentation) than in the original paper (+0.105). This discrepancy may be explained by the baseline already achieving a high F1 on FRII (0.923 vs 0.858 in the original): a network that already separates FRII well has limited room for further improvement, whereas the original paper starts from a weaker baseline and can benefit more from the additional training data.

4 Conclusion

In conclusion, the replication study confirms the main qualitative claims of Brand et al.:

1. Rotational preprocessing improves classification performance over a normalized baseline.
2. Augmentation yields the best Macro F1 at the cost of longer training.
3. Standardization provides an accuracy/training time trade-off that makes it favorable for

resource-constrained applications.

The magnitudes of these effects, however, are strongly influenced by the expressive capacity of the underlying model. This suggests that the conclusions of the original paper are robust to the specific implementation, but that the quantitative benefit of rotational preprocessing should be re-evaluated whenever the network architecture changes substantially. A systematic investigation of this dependency, that is, varying the architecture while keeping the preprocessing fixed, would be an interesting direction for future work.

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